

The Digital Business Marketplace

Catalyst Vision **AFG Perspective**

May 2, 2019



Agile Fractal Grid Honored for “Smart City at the Edge”

Nice, France — June 3, 2017 — Agile Fractal Grid announced today that it was selected as a winner in [TM Forum’s Excellence Awards 2017](#).

Agile Fractal Grid was recognized as the “Smart City Innovator of the Year” for its Smart City at the Edge activities and contributions.



**We were selected
by 640 of our peers in the
telecommunications industry
around the world.**

**This has not gone unnoticed
by the electric power industry.**

The Agile Fractal Grid has two movements.

**Generative
Unfolding**



**Agile
Fractal
Grid**



**Sustaining
Operations**

Innovation Platform

*Community-based
Economic Development*

- *Digital Services*
- *Micromanufacturing*
- *The Creative Arts*
- *Venture Capital*

Infrastructure Services

**We work with local network
providers to add useful
applications to their
available facilities.**

Service Platform

*Shared Multi-Purpose
Orchestration Platform and
Operational Support Services*

**Cellular Service
Companies**

**Internet Service
Providers**

**Electric Utilities –
Coops & Municipals**

**Cloud Computing
Companies**

**Digital Marketplace Platform
Retail Partners**

**Marketplaces:
Energy Services
Public Safety Services
*Community Services***

Potential Range of Services

✓ *Digital Marketplace Portal*

- **Solution as a Service**
- Datacenter as a Service
- Supply Chain Support

✓ *Wholesale Communications*

- **5G IoT Wireless Access**
- Small Cell Cellular Coverage
- Streaming and On Demand Content
- Transportation Systems
- Precision Agriculture
- Sports Venues
- Drone Control

✓ *Electric Power*

- **Microgrid as a Service**
- Virtual Power Pool
- Smart Buildings
- Precision Measurement

✓ *Public Safety as a Service*

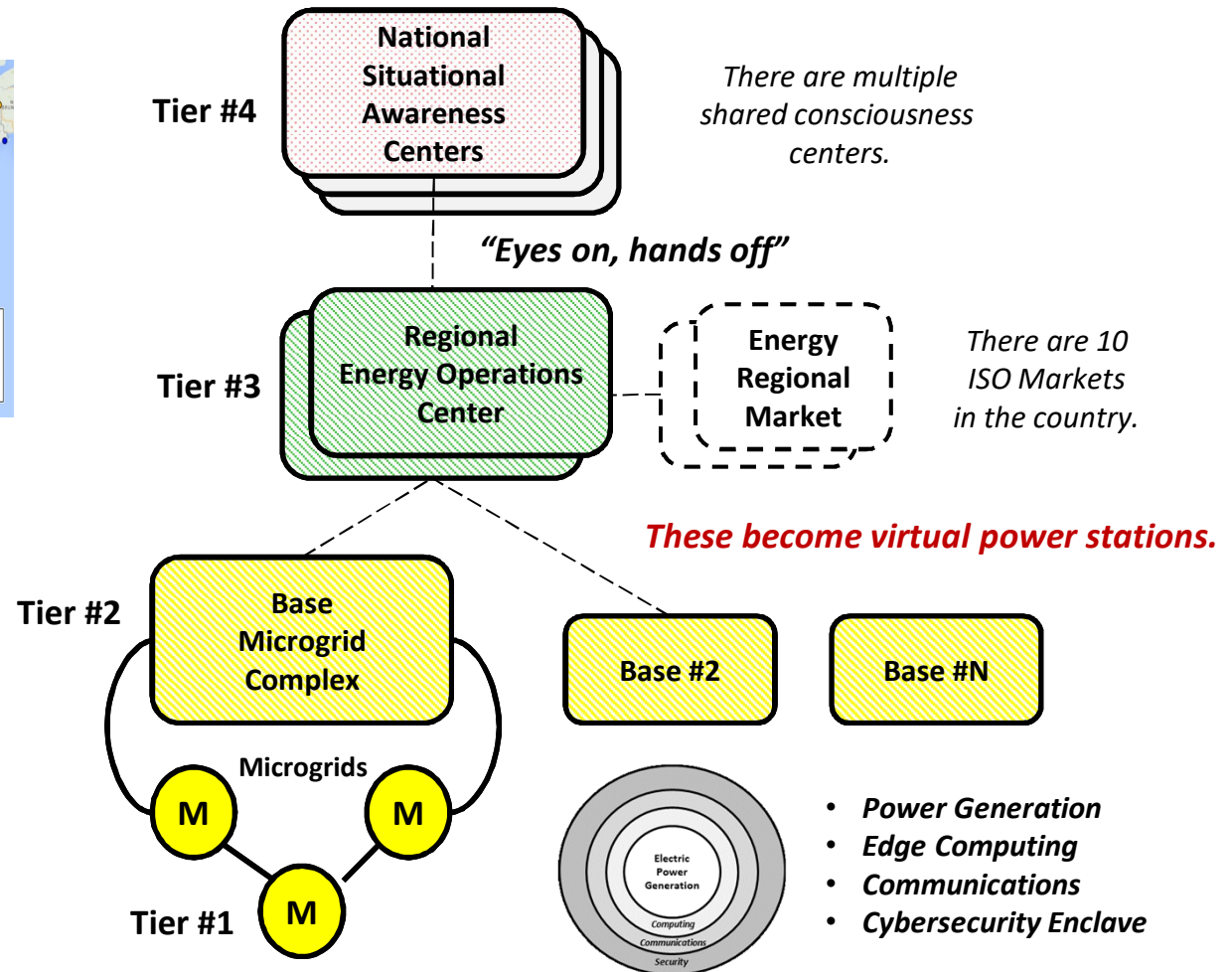
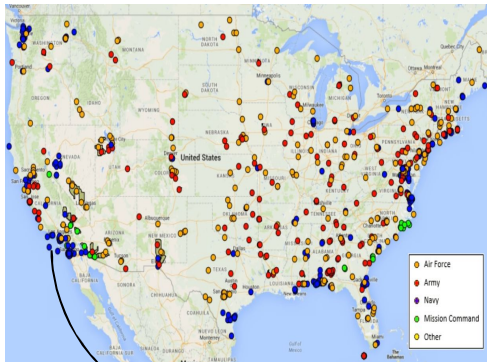
- **Public Safety Interconnect**
- Emergency Operations Center
- Medical Center Integration
- School Safety
- NG E-911

The New Agora

Our platform of platforms allows you to set up your own businesses.

Military Base Fractal Grids Operate Together and Standalone

*There are many military bases
in the country.*



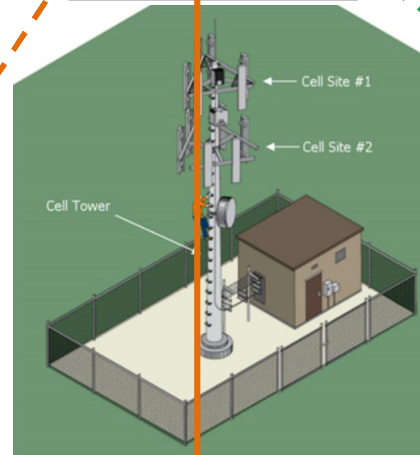
*Each base and each microgrid operates autonomously as needed,
but operates as a team when connected.*

The Fractal Microgrid Yard



- *Other Fixed Devices*
- *Mobile Pads & Devices*

RAN

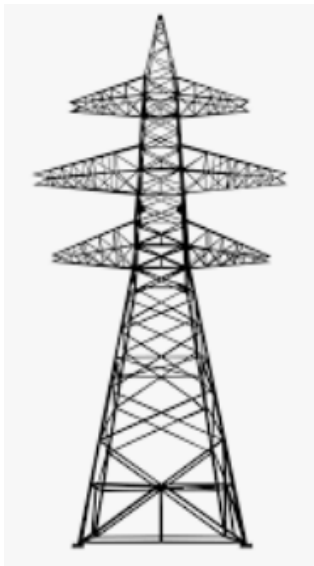


Resilient Path

The Proxy
Field Edge Device

Wired Path

The Pod Cloud

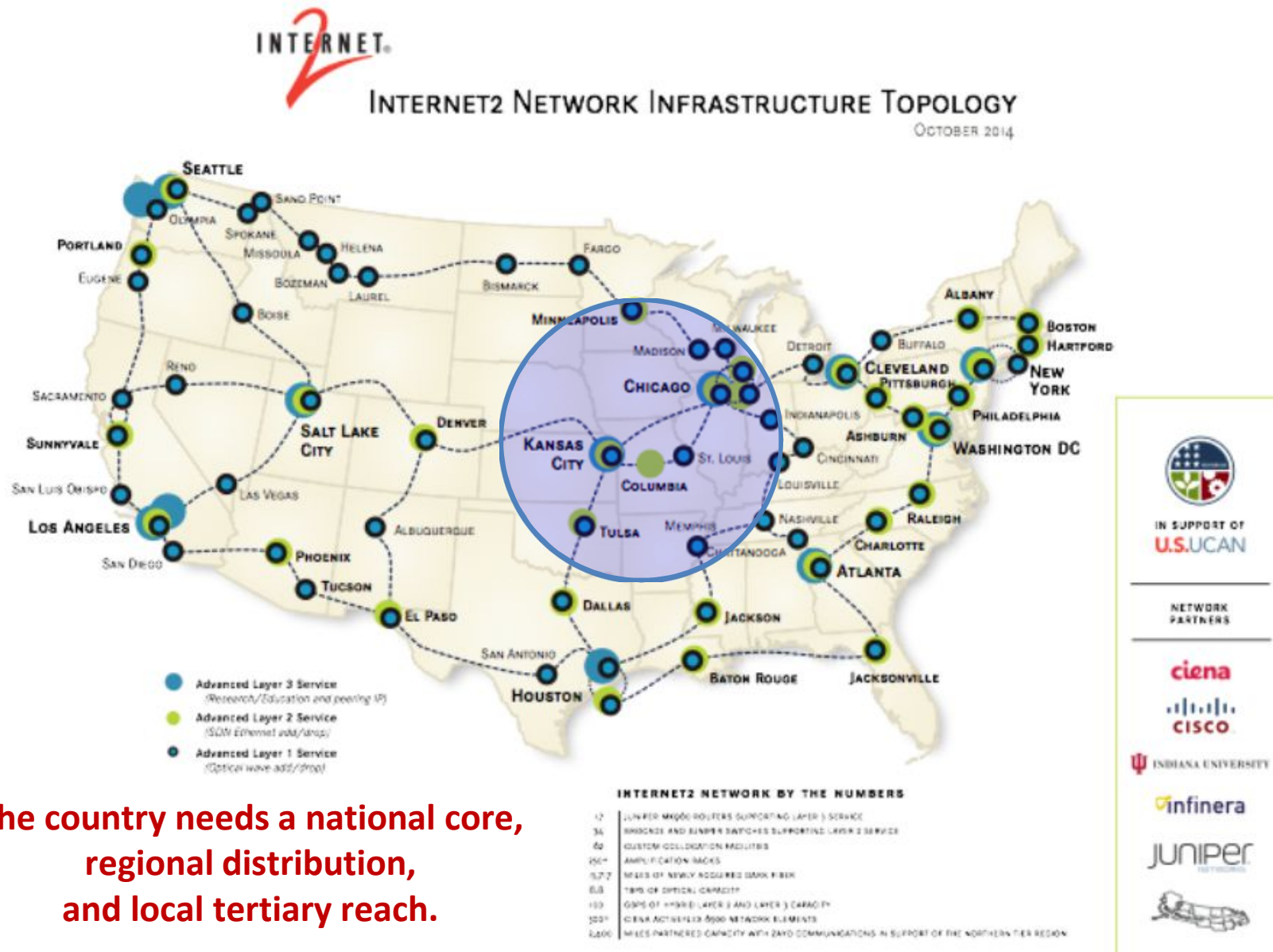


GenTie

- **Generator**
- **Switch**
- **Meter**
- **Synchrophasor**
- *Wind Turbine*
- *PV Array*
- *Battery*
- *Inverter*

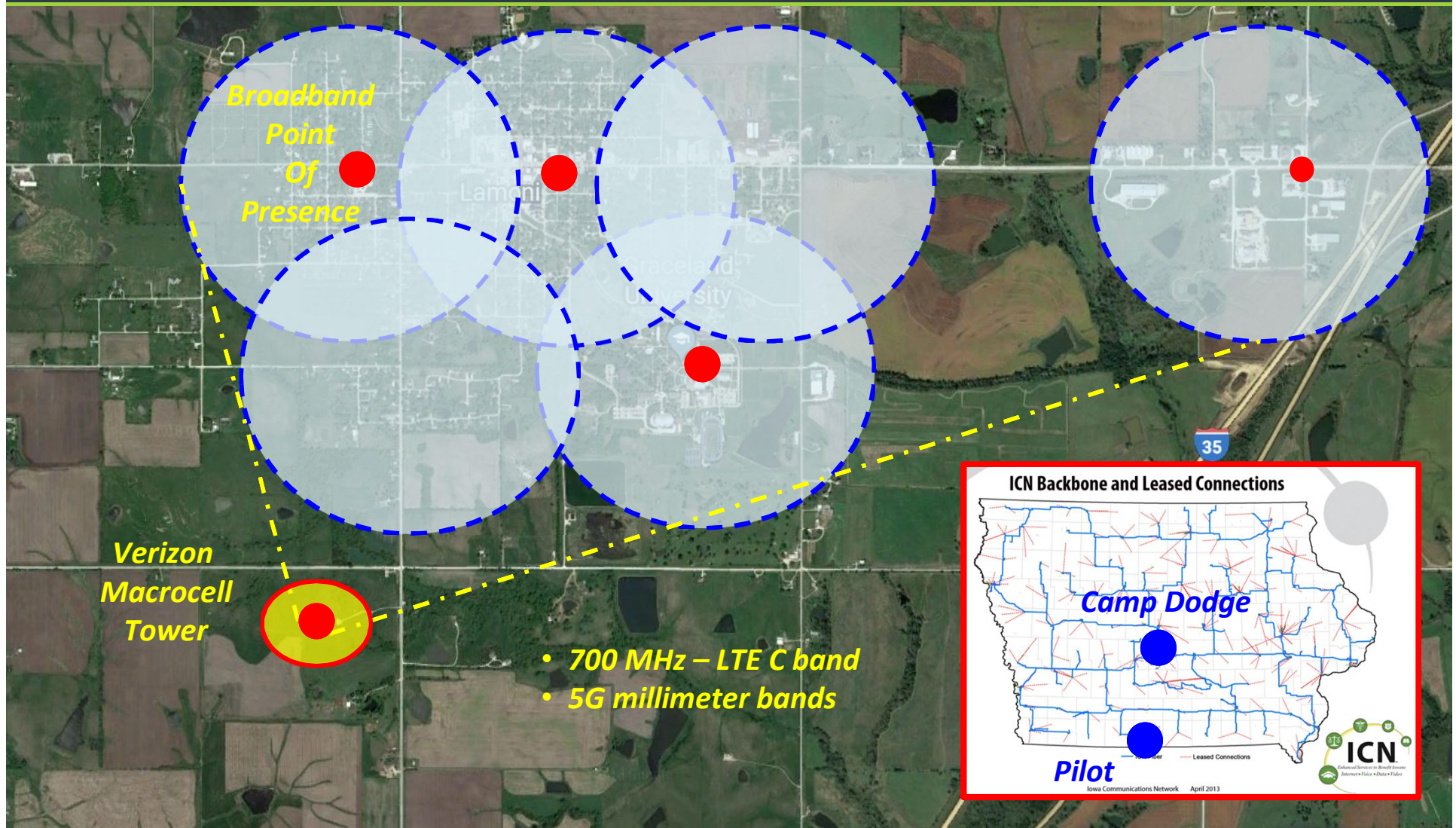
- *"Core" Microservices*
- *Security Expansion Board*
- *Dragos Filter*
- *4G and 5G HetNet Gateways*
- *Optimizer*
- *Microgrid Controller with distributed SCADA*
- *SEL SDN switches*
 - *Modbus*
 - *Profinet*
 - *DDS*
 - *OpenFMB*

Our national core will leverage Internet2.



The country needs a national core,
regional distribution,
and local tertiary reach.

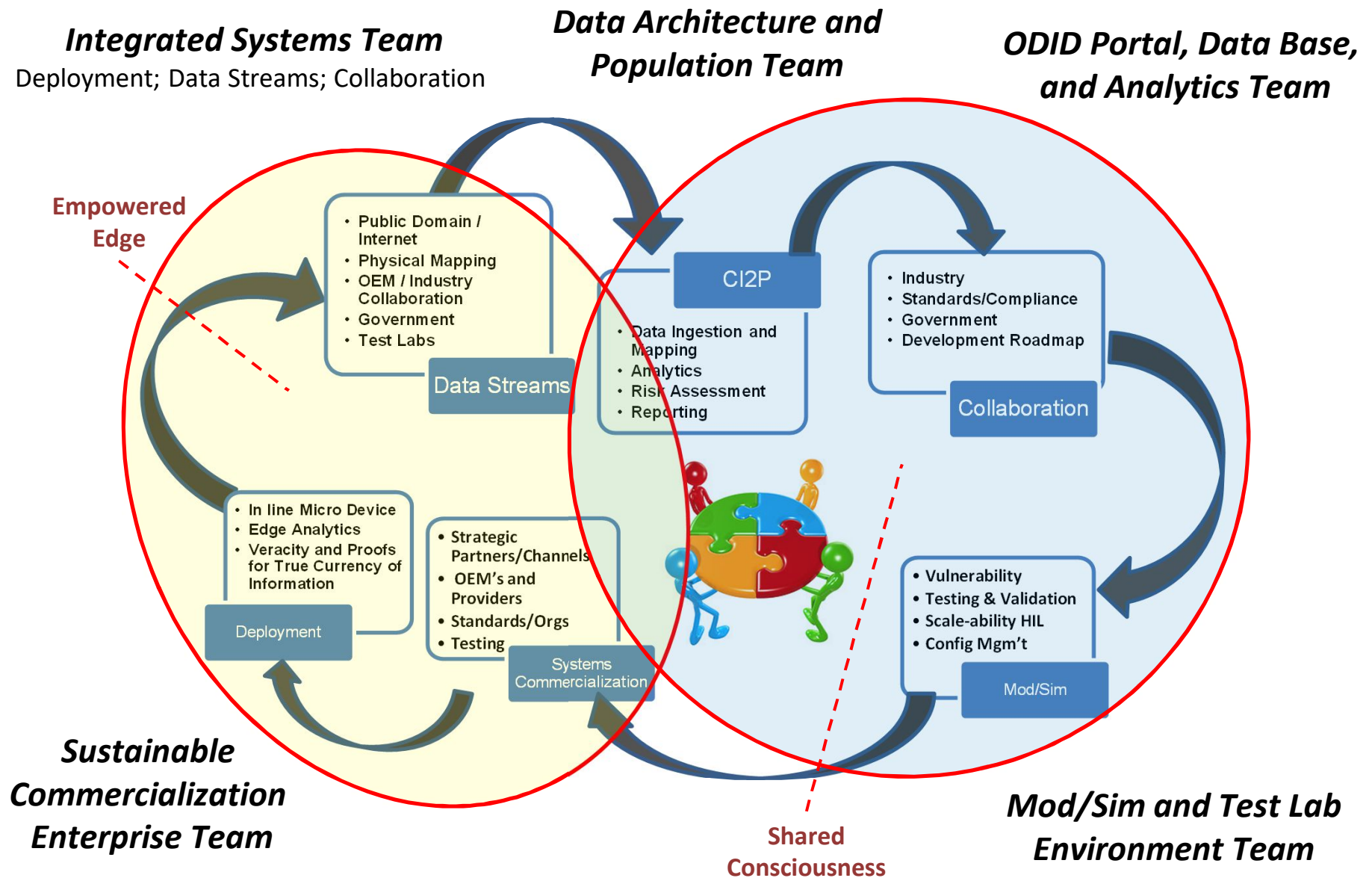
The pilot needs to be a very controlled environment.



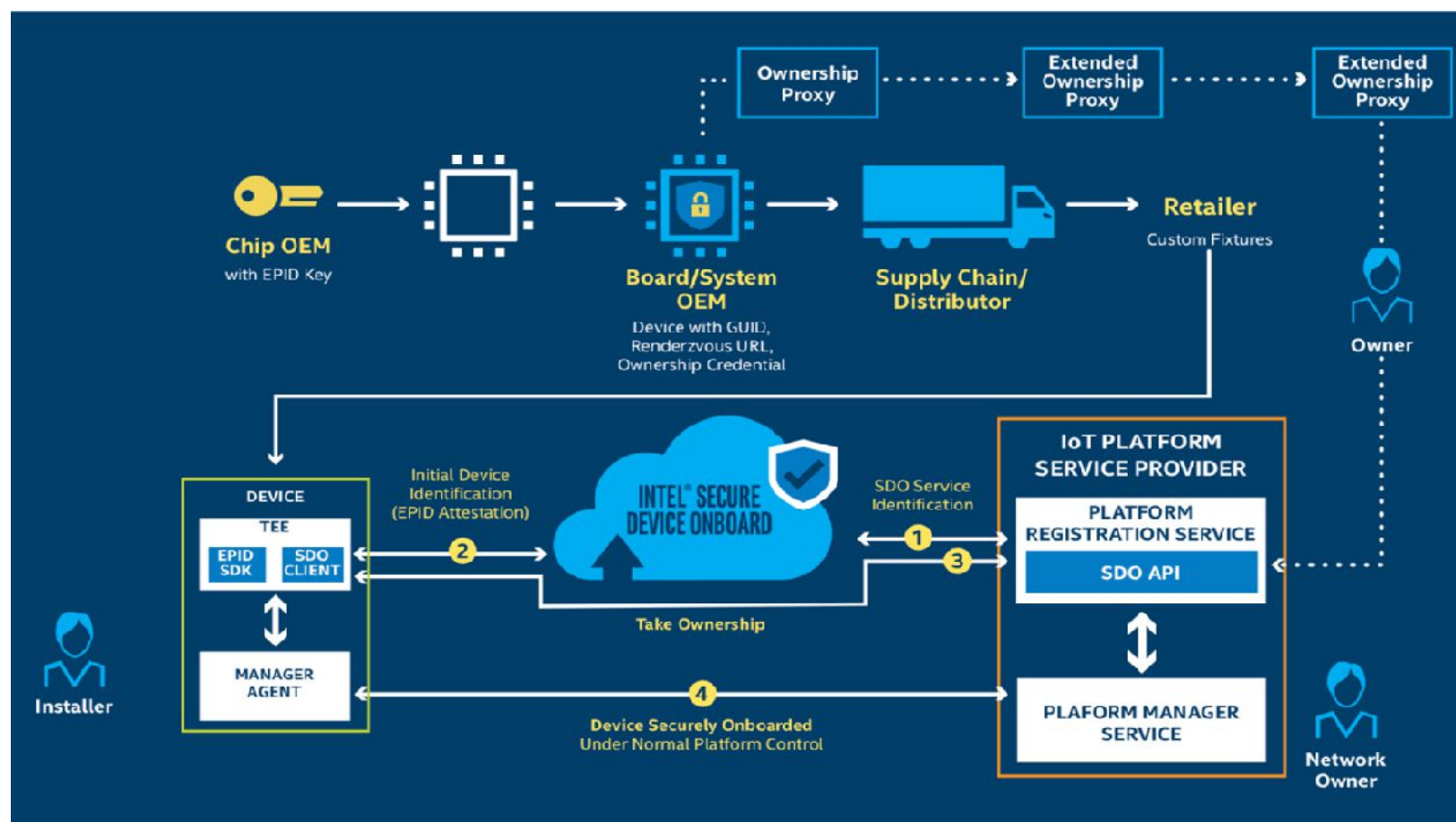
The Iowa Civilian Pilot

- Public Safety
- Commercial Networks
- Tech-You-Topia

The Reinforcing Security Loop

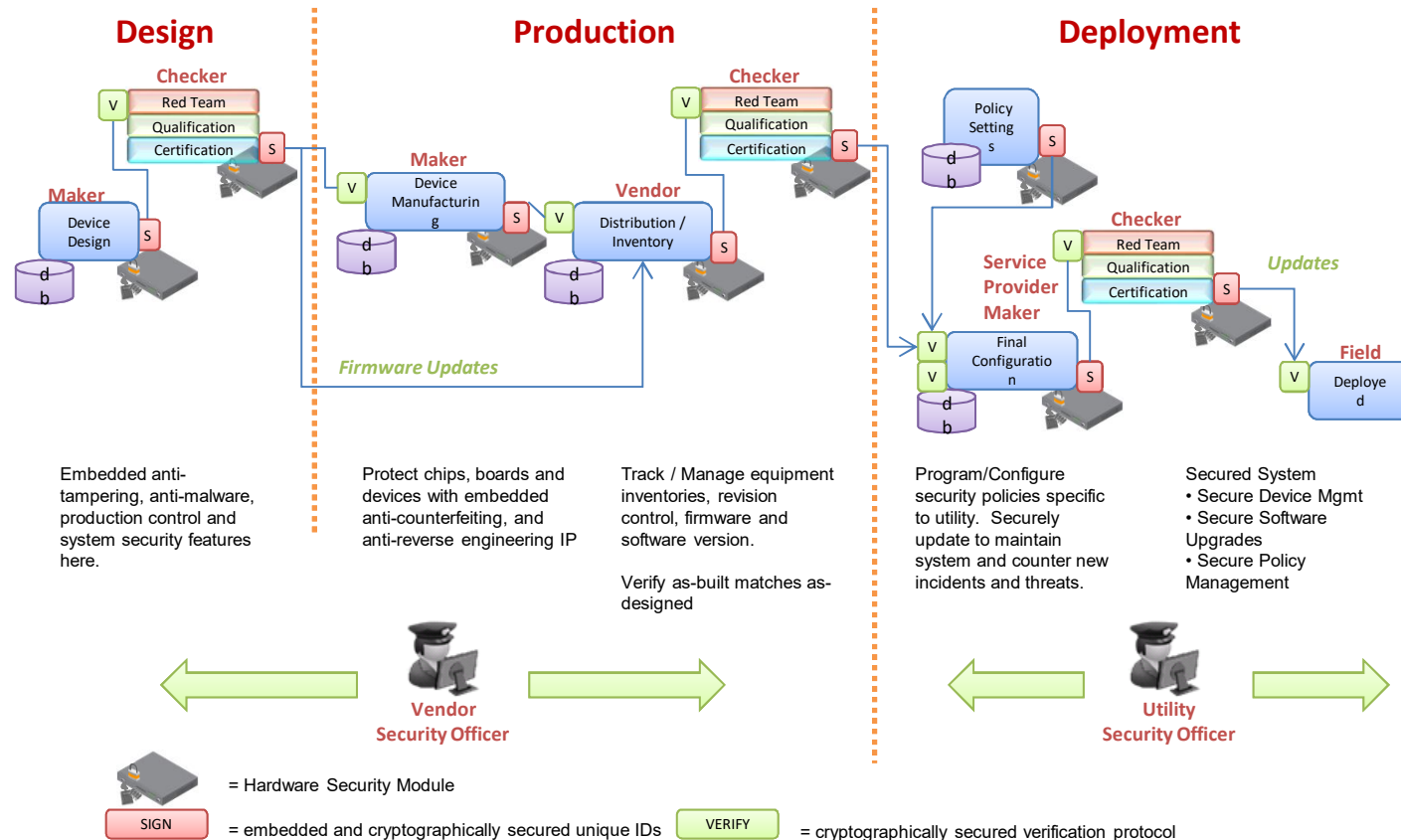


Part of the Catalyst Concept Toward a Secure Supply Chain for the IoT.



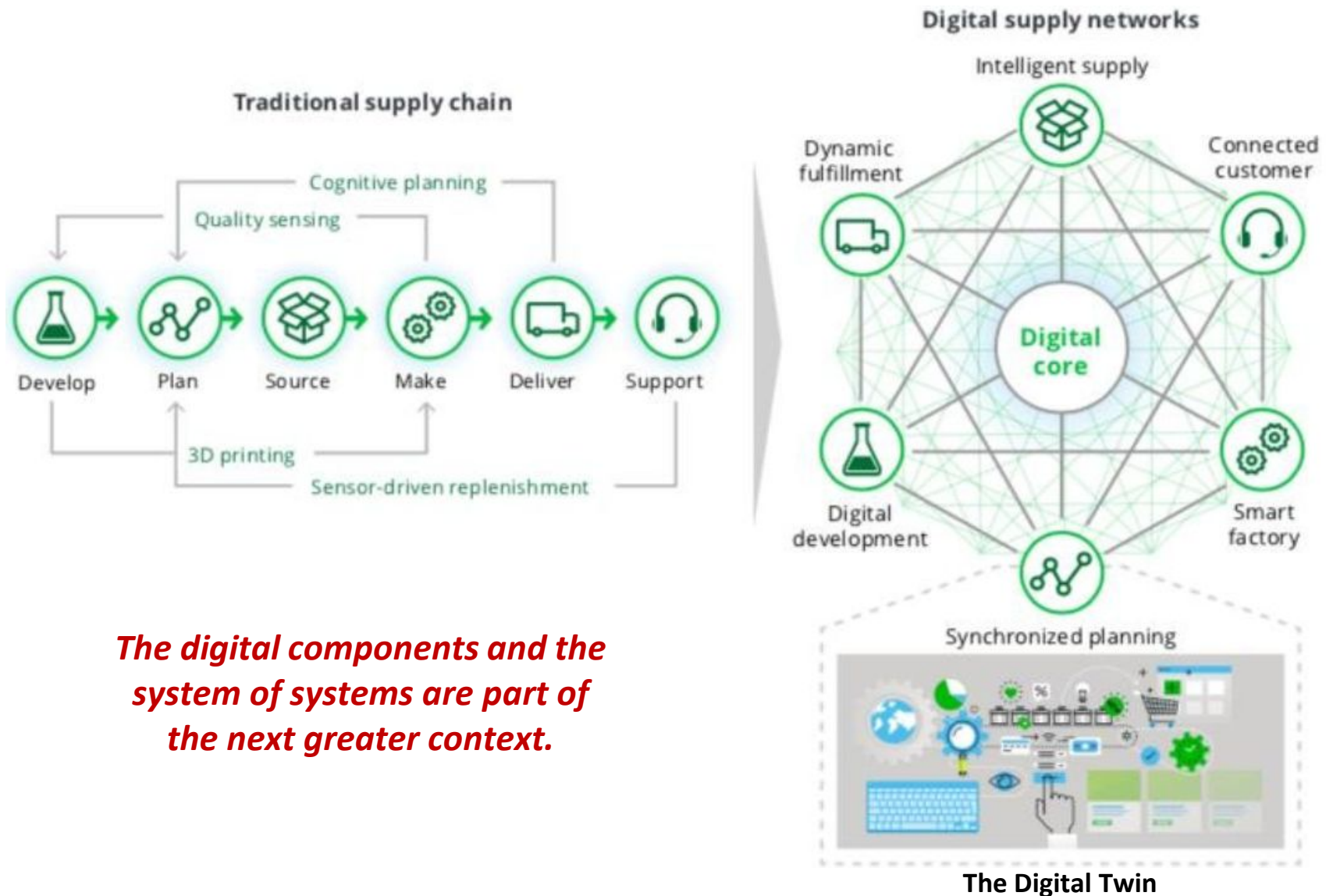
*The Catalyst is a useful lean development step
in the rollout of the secure supply chain in the military and critical infrastructure.*

Constructing a Supply “Chain of Trust”



We will leverage the Catalyst as an initial iteration in the rollout of the actual secure supply chain for our pods and proxies.

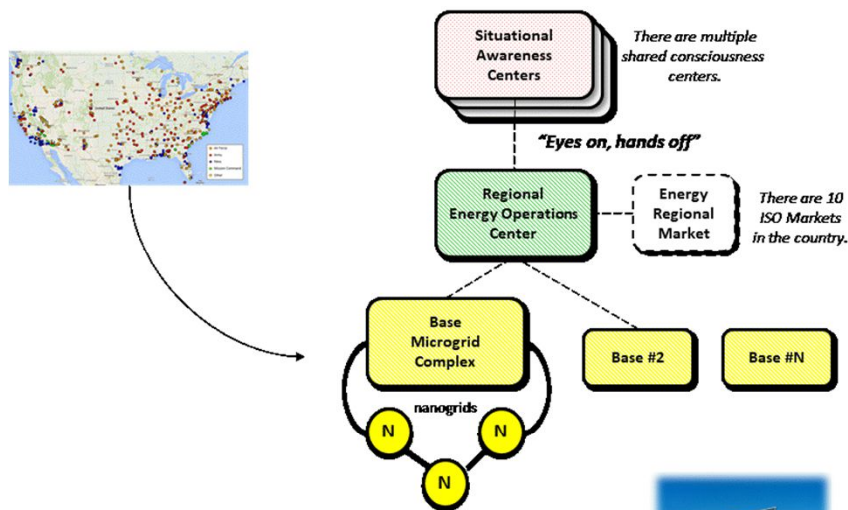
The supply chain for our new pod elements may start to look like more of a digital supply network.



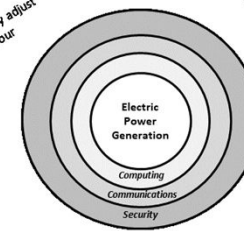


The User Stories

Power Utility Tenant Related User Stories



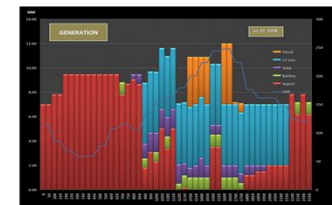
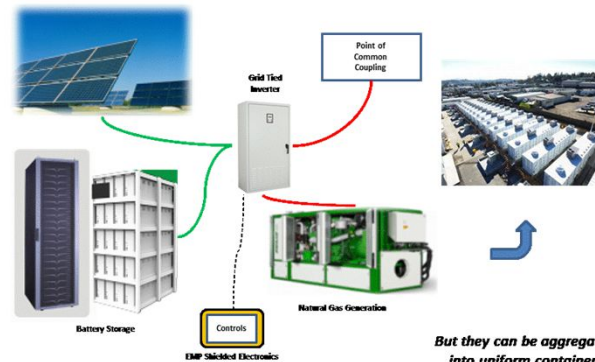
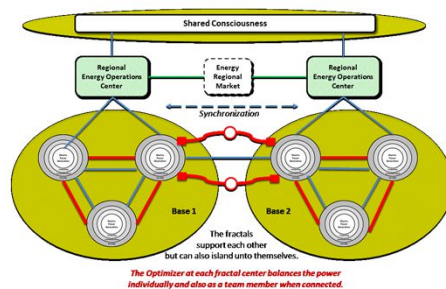
Agile
We must be able to change quickly while we are under attack. But we must also be able to rapidly adjust as the situation evolves in our new networked world.



Grid
This is electricity. It is fundamental to all activities serving mankind.

Fractal
Use simple repeatable patterns that form larger patterns with rhythmic scope, but can operate as independent resilient units but form a reinforcing team of teams at magnificent scale.

Each fractal can launch on its own without waiting for the big system to be complete.



Power Utility Tenant Related User Stories

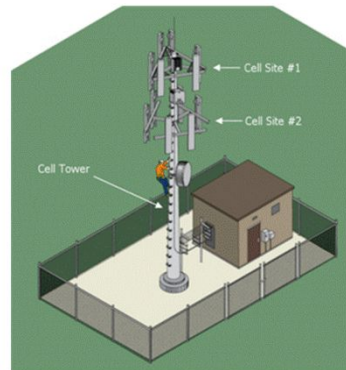
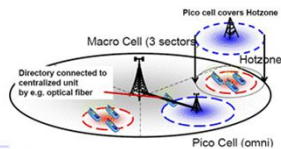
- As a utility power distributor, I want to consume power stability and resilience services that will protect my systems from all internal and external risks, such as the need:
 - to analyse the various data feeds so that my system will automatically isolate my substation in 4 milliseconds, so that the systems can automatically deliver efficient power at all times to my customers
 - to analyse the various data feeds so that my system will automatically reconnect my substation to the main grid, so that the systems can automatically deliver efficient power at all times to my customers
- As the owner of the utility infrastructure, I want to connect up all my transformers with optic fibre so that I can collect all the performance data in real time for substation processing
- As a utility power distributor, I want to consume IIoT services, and deploy sensors throughout the grid, to enable me to collect the data necessary to provide visibility into the state of this complex system

Access Communications Tenant Related User Stories

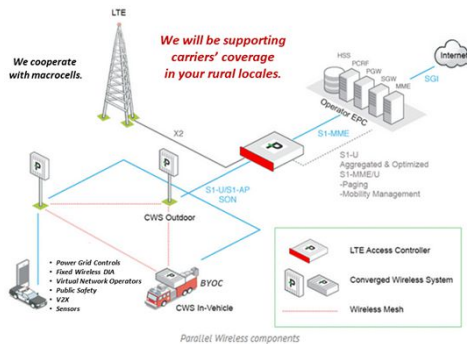


Example of HetNet deployment (Macro + Pico)

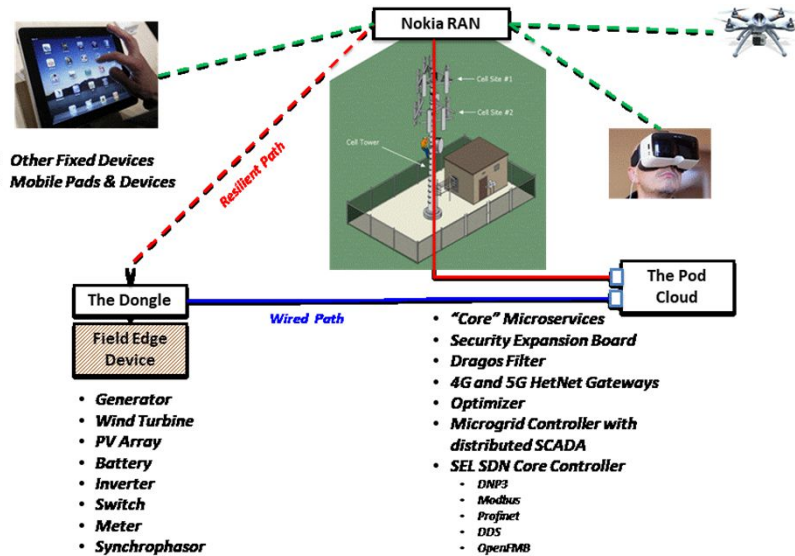
- Macro Cell + Pico Cell
- Pico cells (small cells) are deployed over Hotzones (area with a high concentration of UEs) to
- Transmission power of UEs and base stations can also be reduced. More radio resources can be allocated to a single UE.
- Spectrum efficiency per m² would be improved.



Because of the access to power and computing, the pods make excellent micro and pico cell base stations for the immediate area.



Parallel Wireless components

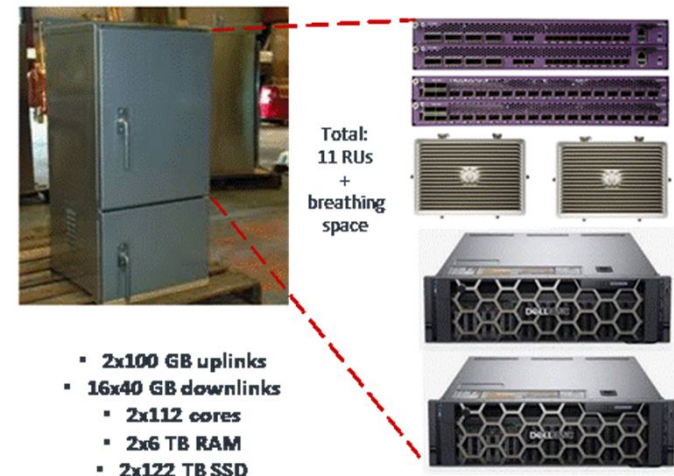
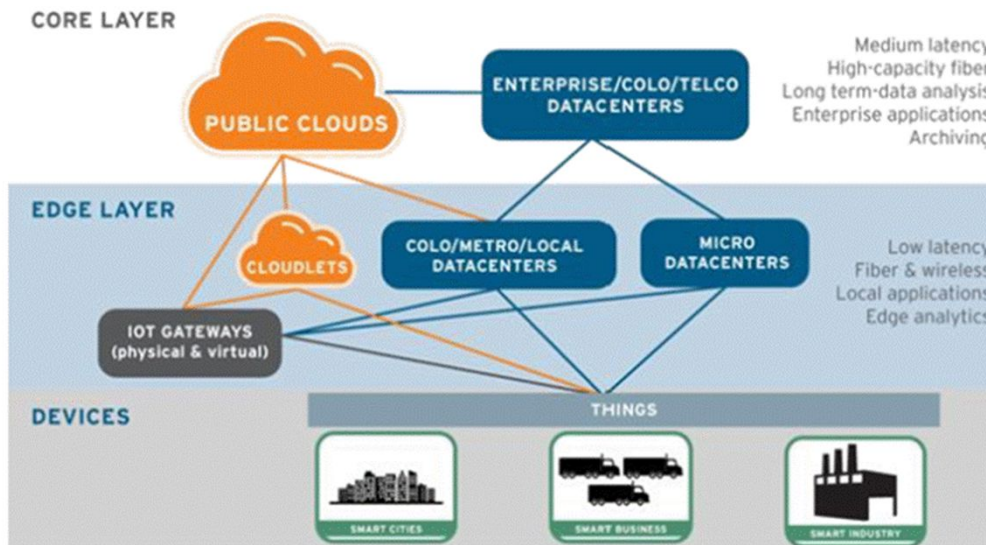


*Each fractal contains its own SDN core.
4G / 5G coverage
MORAN Access Services*

Access Communications Tenant Related User Stories

- As a communications service provider, I want to rent the use of spare optic fibre capacity so that I can expand my geographic coverage and offer 4G and 5G fixed and mobile services to new customers
- As a communications service provider, I want to offer differentiated services which will enable premium pricing to reflect the value points associated with the network slicing and SLA capabilities of 5G.
- As a communication service provider, I want to be able to offer my services to be consumed in conjunction with any adjacent partner's offerings for consumption by other 3rd parties to deliver new business models so that I can establish new revenue streams

Computer Resources Tenant Related User Stories



A computing pod is an environmentally hardened tactical edge computing node.



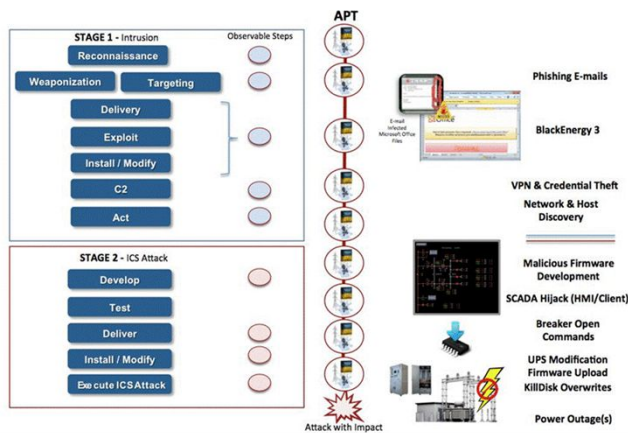
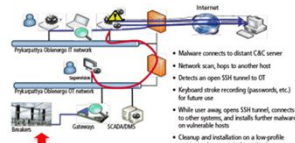
Computer Resources Tenant Related User Stories

- As a provider of computer resources, I want to provide systems that will efficiently scale to support the processing demands of utility power operators, so that the critical 4 millisecond decisions can be achieved
- As a provider of computer resources, I want to provide cloud facilities located at the edge in order to:
 - Be sensitive to the latency requirements for the gracefulness of a continuous operation locally
 - Capture and process data that may be too voluminous to send to a central repository
 - Provide for resilience and continued operation in case of infrastructure failure
- As a provider of computer resources, I want to provide many new edge cloud locations which will appear each and every day in the steady state
- As a provider of computer resources, I want to provide Continuous Integration and Continuous Deployment which are needed to maintain currency at each location each hour of the day
- As a provider of computer resources, I want to have established multiple sites which are required for synchronization and business operation continuance in the face of outages

Cybersecurity Tenant Related User Stories

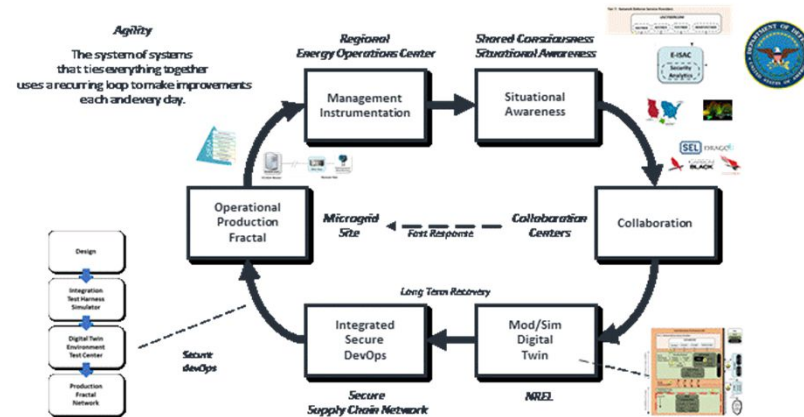


The cyber attack on the Ukraine is a sign of what is to come.



You need to be able to check all the boxes to withstand incursions that use a combined IT/OT strategy.

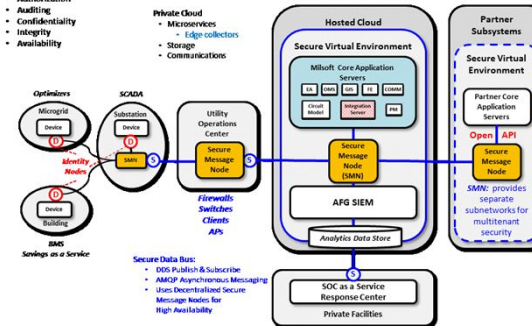
The recurring loop is the mature answer to OT security.



Security Fabric Dimensions:

- Identity Management
- Authentication
- Authorization
- Auditing
- Confidentiality
- Integrity
- Availability

The Security Fabric simplifies end-to-end Cybersecurity.



Cybersecurity Tenant Related User Stories

- As a utility power distributor, I want to consume cybersecurity and resilience services that will protect my systems from all internal, external, physical and cyber risks, so that the systems can automatically deliver efficient power at all times without interruption to my customers
- As a utility power distributor, I have Operations Technology (OT) which requires additional considerations over and above Information Technology (IT). I will need behavioural and systems analytics operating in real-time to deal with contemporary threats
- As a utility power distributor, I note that all of the usual security concepts continue to apply, but as they occur in different locations in IT / OT combined operations, I will need to employ additional services and solutions to ensure that my systems can automatically deliver efficient power at all times without interruption to my customers
- As a utility power distributor, I need to secure end-to-end systems of systems - there must be security at each edge, at the cloud operations point, at partner operations points, and the local service assurance points – and also the shared consciousness centres that support cybersecurity monitoring and response, network operations, systems operations, and electric power operations – and also the end-to-end interconnection technique – so that my systems can automatically deliver efficient power at all times without interruption to my customers

Questions and Answers...



